National University

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Subject: CV Assignment 1 Report

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Abstract

This report presents two computer-vision tasks. The first implements Otsu's thresholding algorithm from scratch on a 16-level quantized grayscale image, deriving the optimal threshold by maximizing between-class variance / minimizing within-class variance, and validating it against OpenCV's built-in Otsu implementation. The second applies intensity-transformation techniques (logarithmic, power-law/gamma, and inverse-logarithmic) to enhance a low-contrast chest X-ray image, comparing the results quantitatively using mean intensity, standard deviation, and dark-region pixel distribution to identify the best enhancement.

Overview:

Image thresholding and intensity enhancement are core preprocessing steps in computer vision. This assignment covers two related problems:

- Question 1: Implement Otsu's automatic thresholding method manually on a 16-level (0–15) quantized grayscale image and segment it into foreground/background.
- Question 2: Apply and compare intensity-transformation functions (log, power-law, inverse-log) to enhance a chest X-ray image with poor contrast.

Question 1 — Otsu's Thresholding

1. Introduction

Otsu's method is a classical automatic thresholding technique that separates an image into foreground and background by finding the threshold that maximizes the between-class variance of pixel intensities. This task implements the algorithm manually on a 16-level quantized grayscale image and validates it against OpenCV's built-in Otsu implementation.

2. Methodology

- Loaded the input image and converted it to grayscale (shape 558×556).
- Quantized intensities from 0–255 down to 16 levels (0–15) by scaling and rounding.
- Built the normalized histogram $p(i)$ over the 16 levels; verified $\Sigma p(i) = 1$.
- For each candidate threshold T (0–15), computed class weights ($w1$, $w2$), class means ($\mu 1$, $\mu 2$), class variances, within-class variance σW^2, and between-class variance σB^2.
- Selected the optimal threshold T* by maximizing σB^2 and independently confirmed it by minimizing σW^2.
- Segmented the image using $g(x,y) = 1$ if $f(x,y) > T^*$, else 0.
- Compared the manual result against OpenCV's cv2.threshold(..., THRESH_OTSU).

3. Results

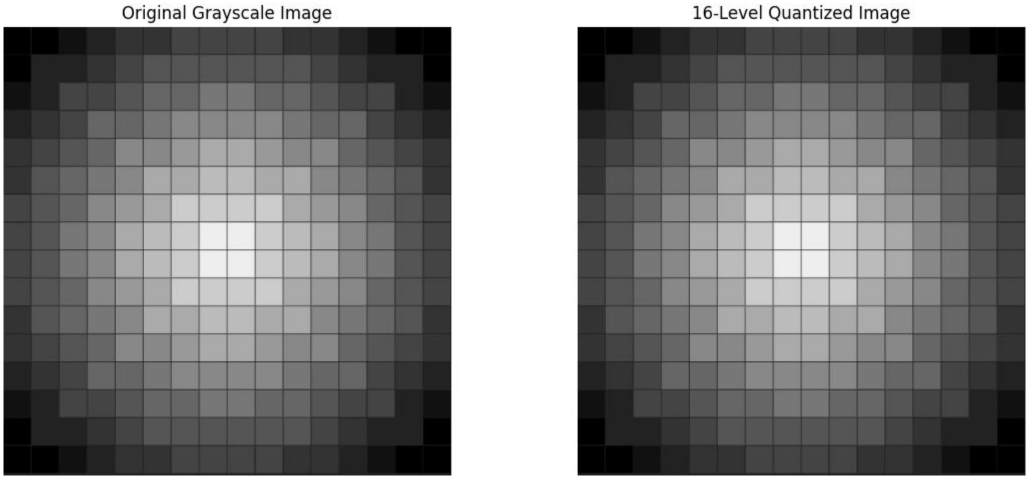

Figure 1: Original grayscale image vs. 16-level quantized image.

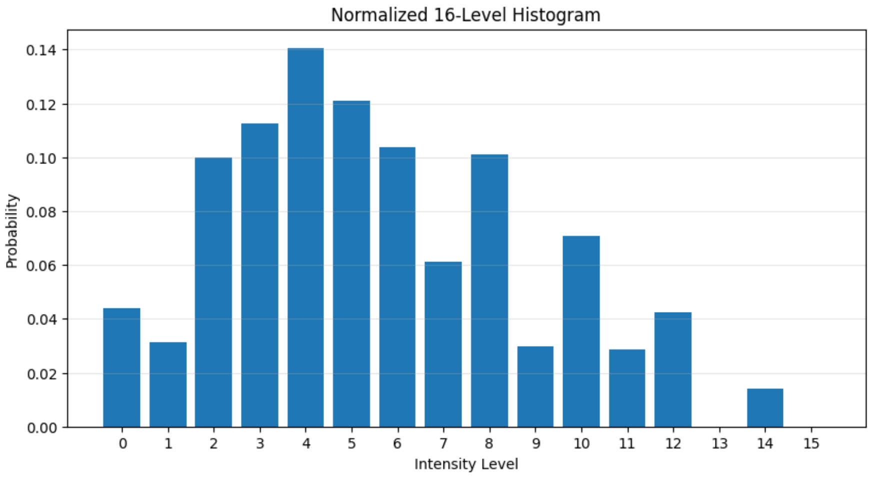

Figure 2: Normalized 16-level histogram.

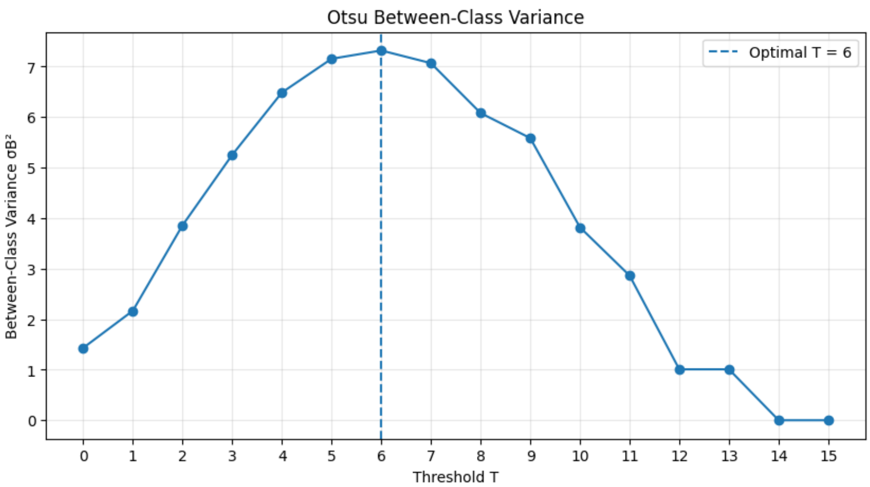

Figure 3: Between-class variance σB^2 vs. threshold T (optimum at $T = 6$).

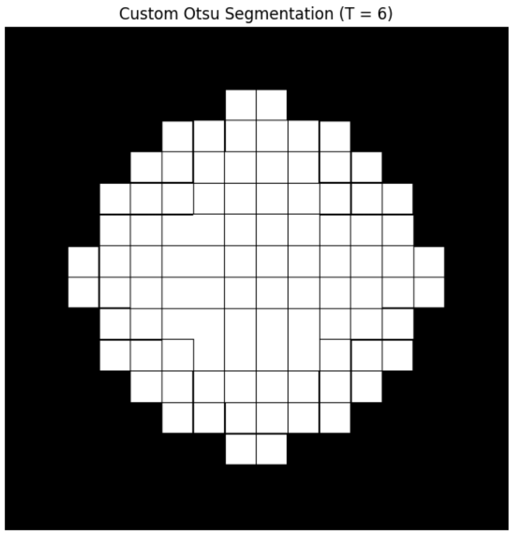

Figure 4: Binary segmentation using the custom Otsu threshold (T = 6).

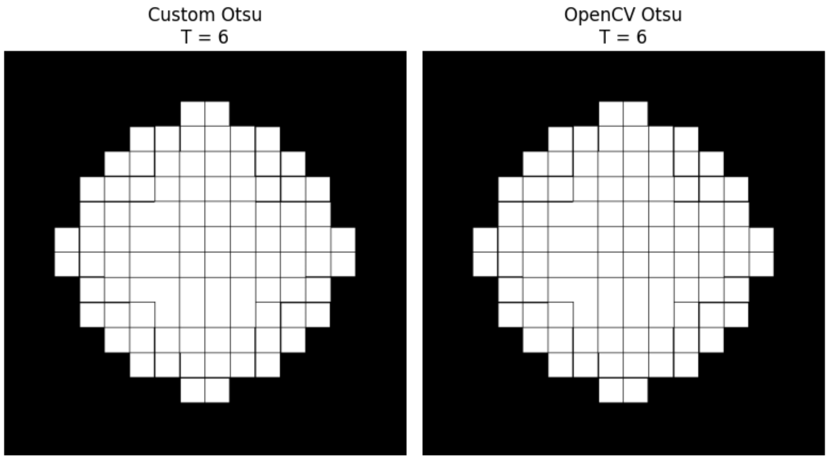

Figure 5: Custom Otsu output vs. OpenCV Otsu output (T = 6, pixel-identical).

Metric	**Result**
Custom Otsu Threshold	6
OpenCV Otsu Threshold	6
Threshold Exact Match	True
Binary Output Exact Match	True
Matching Pixel Percentage	100.00%

4. Discussion

The custom implementation reached the identical optimal threshold ($T^* = 6$) as OpenCV's built-in Otsu method, with a 100% pixel-level match between the two binary outputs. The maximum criteria (maximize σB^2) and minimum criteria (minimize σW^2) agreed, confirming the correctness of the manual derivation. At $T = 6$, background covers 65.25% of pixels and foreground 34.75%.

5. Conclusion

- The manually implemented Otsu thresholding algorithm exactly reproduced OpenCV's result ($T^* = 6$, 100% pixel match), validating the between-class/within-class variance derivation.
- The agreement between the maximize-σB^2 and minimize-σW^2 criteria confirms the theoretical equivalence of the two formulations.

Question 2 — Intensity Transformations

1. Introduction

Intensity transformations reshape the pixel value distribution of an image to improve visibility of features that are hard to see in the original. This task applies and compares logarithmic, power-law (gamma), and inverse-logarithmic transformations on a low-contrast chest X-ray image to identify the enhancement that best improves visibility.

2. Methodology

- Loaded the chest X-ray image (1024×1024, grayscale) and computed baseline statistics.
- Normalized intensities to $r \in [0,1]$ using $r = f(x,y)/(L-1)$, $L = 256$.
- Logarithmic transform: $s = \log(1+r) / \log(2)$.
- Power-law (gamma) transform: $s = r^\gamma$, tested for $\gamma = 0.4, 0.6, 1.0, 1.5, 2.0$.
- Inverse-logarithmic / exponential transform: $s = (e^r - 1) / (e - 1)$.
- Compared all transformations using mean, standard deviation, histograms, and the percentage of pixels below dark-intensity thresholds (50, 75, 100).

3. Results

Figure 6: Original chest X-ray image.

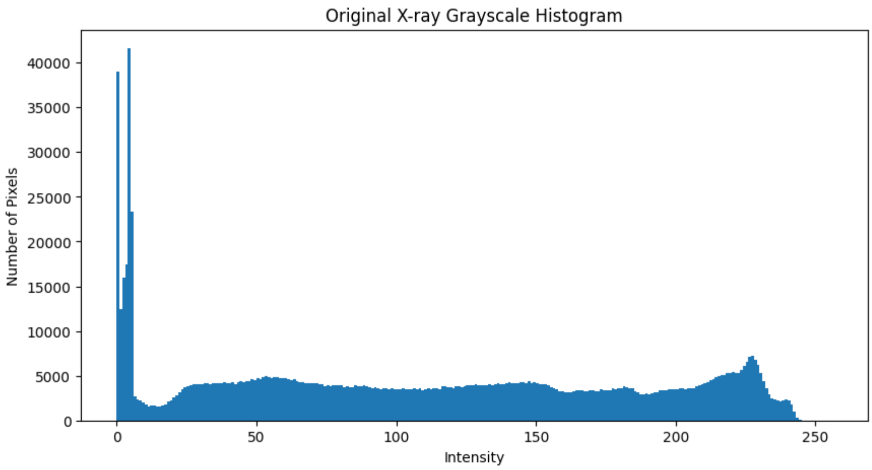

Figure 7: Original image histogram.

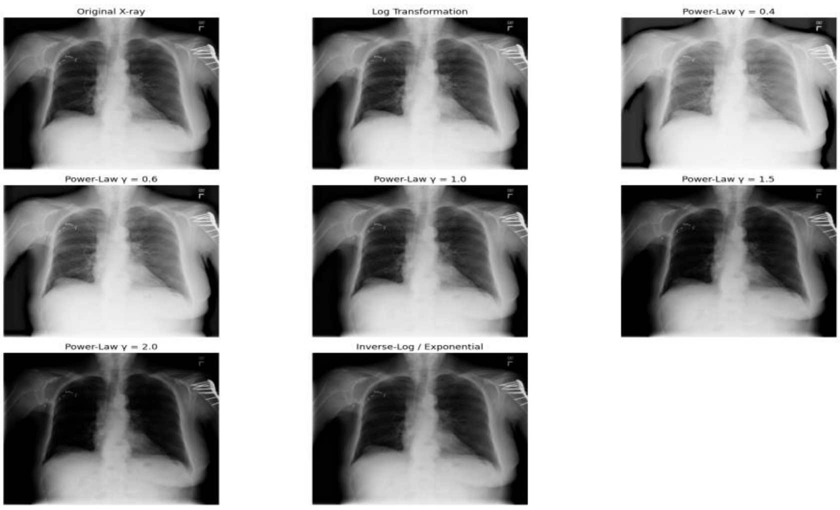

Figure 8: Original vs. all enhancement outputs side by side.

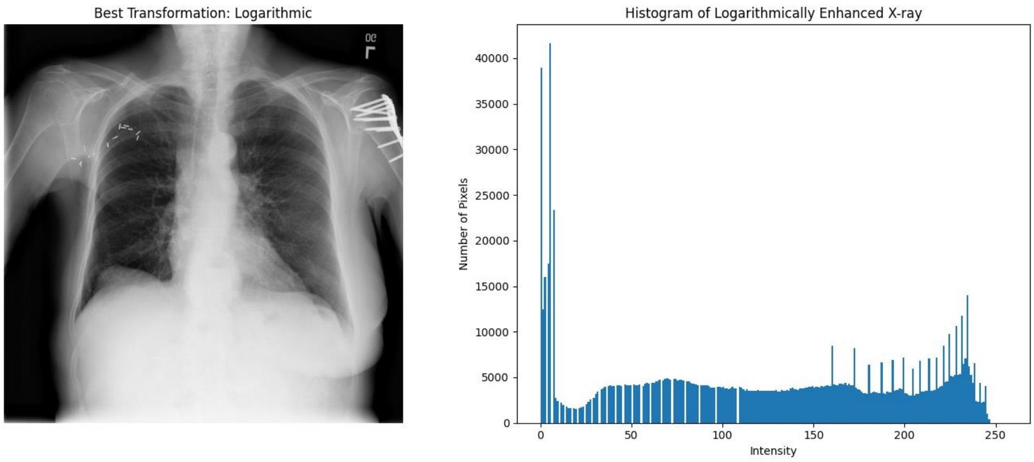

Figure 9: Best transformation (Logarithmic) and its histogram.

Transformation	Min	Max	Mean	Std Dev
Original	0	248	108.64	75.85
Log	0	249	121.87	78.09
Gamma 0.4	0	252	162.07	68.74
Gamma 0.6	0	250	139.05	73.76
Gamma 1.0	0	248	108.64	75.85
Gamma 1.5	0	244	84.19	73.45
Gamma 2.0	0	241	68.45	69.58
Inverse Log	0	244	88.63	70.95

4. Discussion

Gamma = 0.4 produced the brightest output (mean 162.07), while the Logarithmic transformation produced the highest contrast (std. dev. 78.09) and also visibly improved visibility of dark/soft-tissue regions of the X-ray — reducing pixels below intensity 50 from 28.3% (original) to 23.4% — making it the best overall enhancement for this image. Power-law transformations with $\gamma < 1$ brighten the image while $\gamma > 1$ darkens it, giving a tunable enhancement control not available with the fixed log/inverse-log transforms.

5. Conclusion

- Among the intensity-transformation techniques tested, the Logarithmic transformation gave the best enhancement of the low-contrast chest X-ray, improving both brightness and contrast in dark regions.
- Classical pixel-level intensity operations remain effective, interpretable tools for image enhancement.